

Simulation-Based Investigation of Variable Selection

Methods

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Introduction

This project is an investigation of variable selection methods in linear regression using simulations and aims to evaluate how commonly used selection procedures perform when the true underlying model is known. Datasets will be simulated with 20 continuous predictors, five of which will be important and have (true) non-zero effects. Two sample sizes ($N = 250$ and $N = 500$) and several predictor correlation structures will be examined to mimic moderate-sized studies with varying levels of multicollinearity. Across these scenarios, we will compare backward selection based on p-values, AIC, BIC, lasso, and elastic net. Each method will be characterized to determine its ability to identify the true predictors, avoid extraneous variables, and produce unbiased and well-calibrated estimates.

Preliminary Analysis

All analyses will be conducted in R v4.5.2 (2025-10-31 ucrt). Data will be generated using gendata from the hdrm package. Each simulated dataset will include 20 predictors (X1–X20), with X1–X5 assigned non-zero coefficients ranging from $1/6$ to $5/6$ in increments of $1/6$. The remaining predictors will have true coefficients of zero. Errors will follow a standard normal

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distribution. Two sample sizes (250 and 500) will be evaluated. Predictor correlation will be varied using an exchangeable structure with $\rho = 0$ (independent), 0.35 (moderate), and 0.7 (high). Before running the full simulation, a large dataset will be generated to confirm that the simulated data match the intended design, particularly the correlation structure.

For each simulated dataset, five variable selection strategies will be applied: 1. Backward selection using p-values, 2. AIC-based selection, 3. BIC-based selection, 4. Lasso, 5. Elastic net. The following specifications will be used: 1. The full model will be fit using ordinary least squares, and variables will be removed using the step function with a chi-square cut-off corresponding to $\alpha = 0.05$., 2. Stepwise selection will be performed using AIC as the criterion., 3. Stepwise selection will be repeated using BIC., 4. Models will be fit using glmnet with standardized predictors and $\alpha = 1$. Two tuning parameters will be evaluated: lambda.min and lambda.1se from 10-fold cross-validation., 5. Using glmnet with $\alpha = 0.5$, the same two λ choices will be examined. For lasso and elastic net, the set of predictors with non-zero coefficients at the chosen λ will be extracted. Because glmnet does not provide standard errors, the final selected model will be refit using lm to obtain coefficient estimates, confidence intervals, and p-values. Predictors not selected into the model will be treated as non-significant by default.

Each scenario (sample size by correlation level) will be replicated 10,000 times to ensure stable Monte Carlo estimates. Across replications, the following will be summarized: True positive rates (proportion of simulations in which each of the five true predictors is selected), False positive rates (proportion of simulations in which any of the 15 noise predictors is selected), Bias (average difference between estimated and true coefficients for selected vari-

ables), Coverage (proportion of 95% confidence intervals that contain the true coefficient), Type I error (for noise predictors based on significance testing in the refitted models), and Type II error (for true predictors based on significance testing in the refitted models). Because inference after model selection is known to be biased, these summaries will be interpreted descriptively rather than as formal inferential guarantees.

Appendix

Code Directory: <https://github.com/eweible/BIOS6624.git>

File name: P4/Code/P4_Data_Analysis_Plan.Rmd